B

Department of Computer Science and Engineering
Midterm Examination Fall 2024
CSE 321: Operating Systems

Duration: 1 Hour 20 Minutes **Total Marks:** 30

Answer the following questions.
Figures in the right margin indicate marks.

1. CO1

a) After dropping out from college, Steve thought of becoming an entrepreneur. So, he started his company which developed operating systems. His company brought a new operating system to the market. The characteristics of the OS are: **[3]**

- User mode has limited operations.
- User is not concerned in debugging the kernel.
- Large space is taken from main memory when the OS is running.

Figure out the OS structure to be used according to the scenario. **State** proper reasons.

b) As a part of designing your operating system, you finally face the daunting task of designing a scheduler. Your friend suggests a round robin algorithm (**time quantum = 5s**) to implement your scheduler. However, you decide to go with a preemptive version of priority scheduler where the priority of a process is inversely proportional to its burst period. You are given the following arrival burst periods of five processes:

Process	Arrival Time (s)	Burst Time (s)
P1	0	12
P2	1	8
P3	3	6
P4	3	5
P5	7	3

I. **Draw** Gantt Charts for above mentioned algorithms. **[6+6]**
II. **Calculate** average waiting times and turnaround times for both algorithms. **[3+3]**

2. CO2	a) **Find** outputs of the given code:	[6]

```
int main(){
    pid_t a,b;
    double x=15.0;
    double y=11.5;
    double z=5.25;
    a=fork();
    if(a<0){
        printf("fork failed\n");
    }
    else if(a>0){
        printf("Parent Starts\n");
        x=(x+y)-z;
        y=(x-y)+z;
        z=x+y+z;
        wait();
        b=fork();
        if(b<0){
            printf("fork failed\n");
        }
        else if(b==0){
            printf("Another Child Starts\n");
            x=x+y;
            y=y+z;
            z=x+z;
        }
        else{
            wait();
            x=x+y;
            y=y+z;
            z=x+z;
        }
    }
    else{
        printf("Child Starts\n");
        wait();
        x=x*y;
        y=x/y;
        z=(z*y)/x;
    }
    printf("x = %lf\n",x);
    printf("y = %lf\n",y);
    printf("z = %lf\n",z);

    return 0;
}
```

	b) Alex wants to develop such a platform where users from around the world can share information among themselves. His motive is to connect users under a peer to peer network and use a dedicated storage to exchange information. In order to share information users require to send and receive data in the storage among each other. As all users are connected under a peer to peer network therefore, if any user sends data in the storage, it gets broadcasted to other users connected to the network. For his platform Alex is more concerned about functions of peer to peer network over the size of the storage. **Identify** with proper logic which communication method is suitable for Alex's platform?	**[3]**